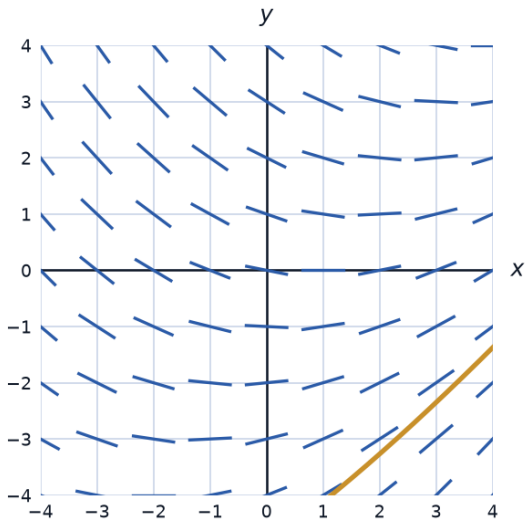
Nonhomogeneous Equations & Undetermined Coefficients

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

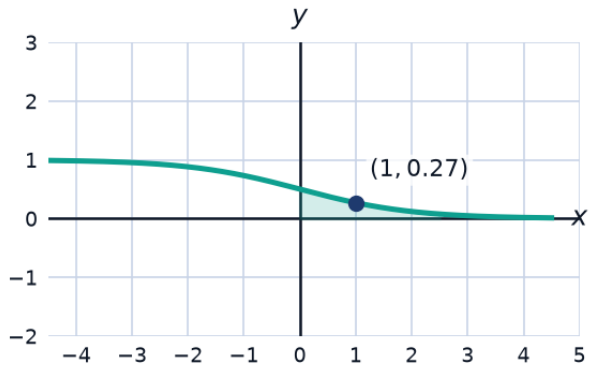
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the differential equation $y'' - 3y' + 2y = f(t)$ with initial conditions $y(0) = 0$ and $y'(0) = 0$. If the forcing function is $f(t) = u(t-2)$, where $u(t)$ is the Heaviside step function, what is the Laplace transform of the solution $Y(s) = \mathcal{L}\{y(t)\}$?



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

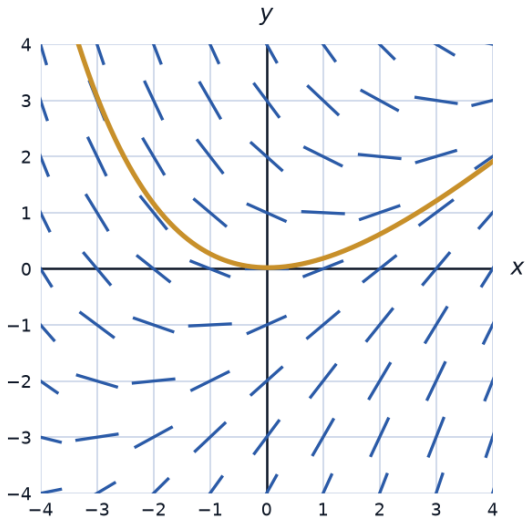
Construct the Green's kernel $G(x, t)$ for $L[y] = y'' - y$, and express the particular solution of $y'' - y = \frac{1}{e^x + 1}$ satisfying $y(0) = 0, y'(0) = 0$ as an explicit definite integral.



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3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

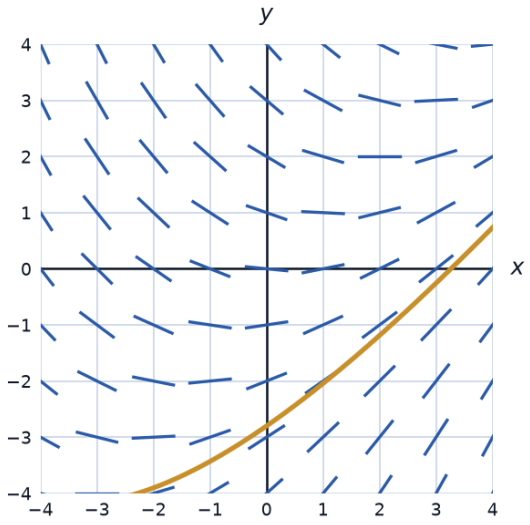
Find a particular solution of $y'' + 4y = \sec(2x)$ for $x \in (-\pi/4, \pi/4)$ using Variation of Parameters.



4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the initial value problem with step forcing using the Laplace transform:

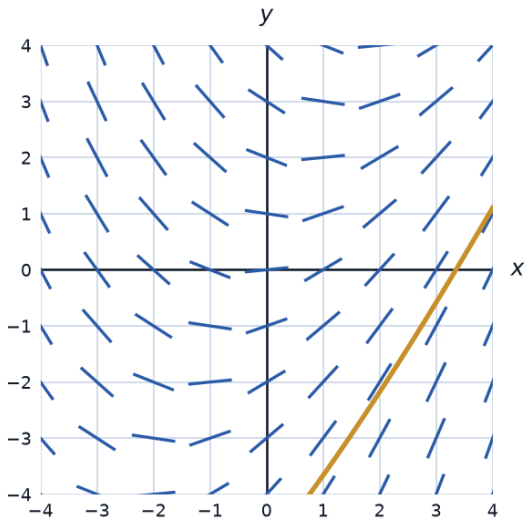
$y'' + 4y = 8u(t-2), \qquad y(0) = 0, \quad y'(0) = 0.$



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5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. You are asked to approximate $y(0.2)$ using Euler's method. If the step size $h = 0.1$ is used, and it is known that the exact solution at $x = 0.2$ is $y(0.2) = 1.24$, determine the value of k such that $y(0.2)_{\text{Euler}} = 1.24 - k$.

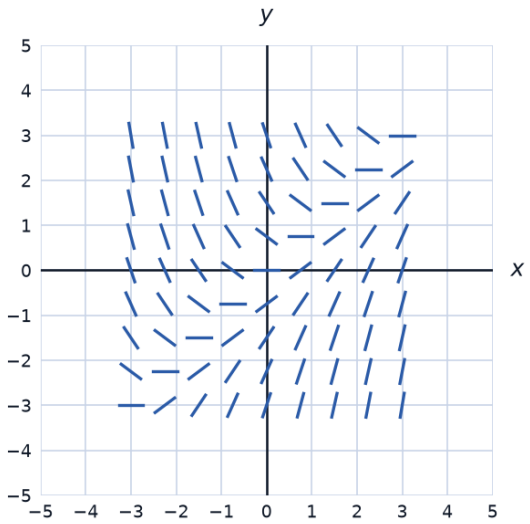


6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the coefficient a_5 of the power series solution $y = \sum a_n x^n$ of $y'' - xy' - y = 0$ with $y(0) = 1$, $y'(0) = 2$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A harvested population evolves according to $\frac{dP}{dt} = P\left(1 - \frac{P}{10}\right) - 2.1$. Determine the two equilibrium populations, identify which is the extinction threshold, and determine whether a starting population $P(0) = 2$ survives or collapses.

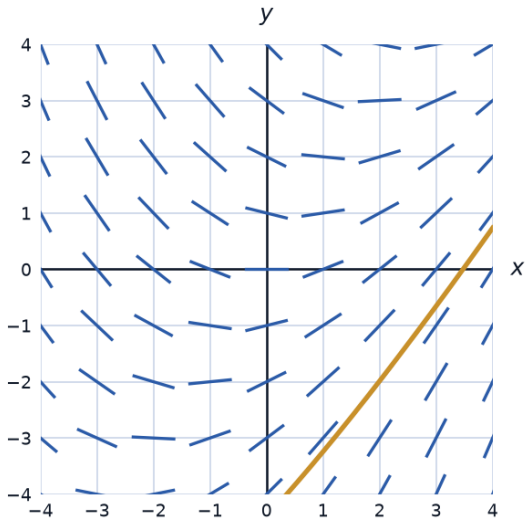


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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve $y'' + 4y' + 4y = 0$ with $y(0) = 2$ and $y'(0) = 1$. Determine the exact time $t > 0$ at which $y(t)$ attains its absolute maximum value.



9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the stiff decay equation $\frac{dy}{dt} = -100y$ with $y(0) = 1$. Determine the stability threshold h_{crit} for Forward Euler, and compute the value of $y(0.1)$ using Backward (Implicit) Euler with step size $h = 0.05$.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Use the Laplace transform and the Convolution Theorem to solve the integral equation $y(t) = t^2 + \int_0^t e^{t-\tau} y(\tau)\, d\tau$.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the initial value problem $\frac{dy}{dx} = x + y$ with $y(0) = 1$. Use the Euler method with step size $h = 0.5$ to approximate $y(1)$.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve $\frac{dy}{dx} = \frac{1+y^2}{x^2-4}$ on $(-2, 2)$ with $y(0) = 1$. At what value of x does $y(x) \to +\infty$?

13 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve $x^2y' + xy = \frac{1}{x^2+1}$ for $x > 0$, with $y(1) = \frac{\pi}{4}$. What is $y(\sqrt{3})$?

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Nonhomogeneous Equations, Undetermined Coefficients & Resonance · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor**

Level 1 · FoundationsOn an AP Calculus BC exam question, a student writes $\int_0^5 (R(t) - D(t))\,dt = 42.18$ as their final response to a question asking for the total amount of water in a tank at $t = 5$, given initial volume 100 gallons. Identify the missing component in the student's solution.

16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations**

Level 3 · Advanced / AP FRQLet $f(x) = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$. Identify the closed-form elementary function that this series represents, and determine its radius of convergence.

17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Variation of Parameters**

Level 2 · MediumOn Section II Part A (Calculator Active), a student is asked for the acceleration vector at $t = 3$ for a particle whose velocity is $\vec{v}(t) = \langle \ln(t^2 + 1), 3\cos(t) \rangle$. What calculator command must be used to find this vector, and what is its 3-decimal numerical value?

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Separable Equations**

[AP Calculus BC Free-Response Questions (FRQs) · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor**

Apply two steps of Heun's method (Improved Euler) with step size $h = 0.1$ to approximate $y(0.2)$ for the IVP $\frac{dy}{dt} = 2ty, \qquad y(0) = 1.$ Compare your approximation with the exact analytical value $y(0.2) = e^{(0.2)^2} = e^{0.04} \approx 1.04081077.$

20 **[Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations**

An RC circuit with resistance $R = 2\,\Omega$ and capacitance $C = 0.5\ \text{F}$ is driven by a delayed step voltage $E(t) = 10u(t - 2)$. With $q(0) = 0$, find the exact charge $q(t)$ for $t \geq 0$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the structure of the general solution to a linear nonhomogeneous equation $L[y] = g(x)$?

a. $y(x) = y_c(x) + y_p(x)$, where y_c is the complementary solution and y_p is any particular solution

b. $y(x) = y_c(x) \cdot y_p(x)$

c. $y(x) = y_p(x)$ only

d. $y(x) = c_1 y_p(x) + c_2 y_c(x)$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find a particular solution y_p to $y'' - 5y' + 6y = 12$.

a. $y_p = 2$

b. $y_p = 12$

c. $y_p = 2x$

d. $y_p = 6$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the trial particular solution y_p for $y'' + 4y = 3x^2 - 1$?

a. $y_p = Ax^2 + Bx + C$

b. $y_p = Ax^2$

c. $y_p = Ax^2 + B$

d. $y_p = Ax^3 + Bx^2$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the correct undetermined coefficients form for $y'' + 9y = 5\cos(2x)$?

a. $y_p = A\cos(2x) + B\sin(2x)$

b. $y_p = A\cos(2x)$

c. $y_p = Ax\cos(2x)$

d. $y_p = A\sin(2x)$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

When must a trial particular solution y_p be multiplied by x (or x^2)?

a. When any term in the trial solution is already a solution of the associated homogeneous equation

b. Whenever the forcing function is a polynomial

c. When the coefficient of y' is negative

d. Only when the right-hand side is zero

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For $y'' - 4y' + 4y = 3e^{2x}$, the homogeneous solution is $y_c = (c_1 + c_2x)e^{2x}$. What is the correct trial form for y_p?

a. $y_p = Ax^2e^{2x}$

b. $y_p = Ae^{2x}$

c. $y_p = Axe^{2x}$

d. $y_p = Ax^3e^{2x}$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What physical phenomenon occurs when an undamped mechanical system is driven at its natural resonance frequency ($y'' + \omega_0^2 y = F_0\cos(\omega_0 t)$)?

a. Resonance, where amplitude grows linearly with time ($y_p = Ct\sin(\omega_0 t)$)

b. Immediate decay to zero

c. Critical damping

d. Chaotic motion

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If y_{p1} solves $L[y] = g_1(x)$ and y_{p2} solves $L[y] = g_2(x)$, what solves $L[y] = 2g_1(x) - 3g_2(x)$?

a. $2y_{p1} - 3y_{p2}$

b. $y_{p1}y_{p2}$

c. $\frac{y_{p1}}{y_{p2}}$

d. $y_{p1} + y_{p2}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Can the Method of Undetermined Coefficients be used to solve $y'' + y = \tan x$?

a. No, because derivatives of $\tan x$ produce infinitely many non-terminating forms; Variation of Parameters is required

b. Yes, with trial solution $A\tan x + B\sec x$

c. Yes, with trial solution $A\sin x/\cos x$

d. Only if x is small

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An undamped oscillator satisfies $y'' + 4y = 3\cos 2t$ (forcing at exactly the natural frequency), $y(0) = 0$, $y'(0) = 0$. Find $y(\pi/4)$.

a. $-3\pi/16$

b. $3\pi/16$

c. $3\pi/8$

d. $\pi/4$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find a particular solution of $y'' - y' - 6y = e^{2x} + 18$.

a. $14e^2x - 3$

b. $-14e^2x - 3$

c. $-14xe^2x - 3$

d. $-14e^2x + 3$

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A plane region R is bounded by the graph of $y = e^{-x^2}$, the x-axis, and the lines $x = 0$ and $x = 1$. A solid is generated by revolving the region R about the line $y = -2$. The cross-sections of the solid perpendicular to the x-axis are squares. The volume of this solid is given by $V = \int_a^b A(x)dx$. (a) Find the outer radius and inner radius of a representative disk if the region were revolved about the line $y = -2$ using the washer method. (b) Set up, but do not evaluate, an integral expression for the volume of the solid generated by revolving the region R about the line $y = -2$ using the washer method. (c) Set up, but do not evaluate, an integral expression for the volume of the solid generated by taking cross-sections perpendicular to the x-axis that are squares. (d) Explain why the integral expression in part (c) is the correct representation for the volume of the solid.

a. The radius of a disk in the washer method is $R(x) = e^{-x^2} + 2$ and $r(x) = 2$.

b. The volume using the washer method is $V = \int_0^1 \pi((e^{-x^2} + 2)^2 - 2^2)dx$.

c. The area of a square cross-section is $A(x) = (e^{-x^2})^2$.

d. The integral expression in part (c) represents the volume because it sums the areas of square cross-sections perpendicular to the x-axis over the interval [0, 1].

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider a region R in the xy-plane bounded by the graphs of $y = \frac{1}{2}x^2$ and $y = 2x$. A solid has its base in region R, and for each value of x in the interval $[0, 4]$, the cross-section perpendicular to the x-axis is an equilateral triangle with side length $s(x) = 2x - \frac{1}{2}x^2$. Write an integral expression that represents the volume of this solid. You do not need to evaluate the integral.

a. $V = \int_0^4 \frac{\sqrt{3}}{4}\left(2x - \frac{1}{2}x^2\right)^2 dx$

b. $V = \int_0^4 \frac{\sqrt{3}}{4}\left(2x - \frac{1}{2}x^2\right)dx$

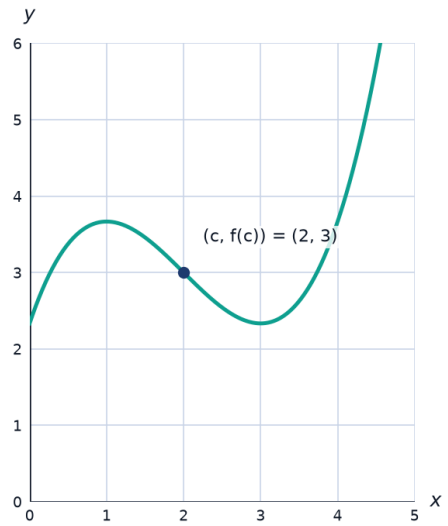
c. $V = \int_0^4 \frac{1}{2}\left(2x - \frac{1}{2}x^2\right)^2 dx$

d. $V = \int_0^4 \frac{\sqrt{3}}{2}\left(2x - \frac{1}{2}x^2\right)dx$

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of a differentiable function f is shown below. The function f has a local maximum at $x = a$ and a local minimum at $x = b$. The tangent line to the graph of f at $x = c$ has the equation $y - f(c) = \frac{1}{2}(x - c)$. Given that $f(c) = 3$ and $f'(c) = \frac{1}{2}$, find the value of c.



a. 4

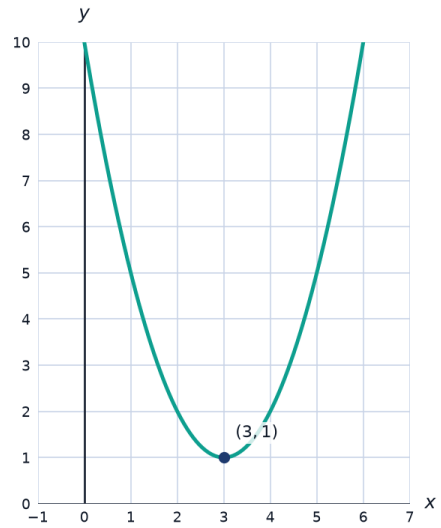
b. 3

c. 2

d. 1

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the function $f(x) = x^2 - 6x + 10$. Find the x-coordinates of the points on the graph of $y = f(x)$ where the tangent line is horizontal.



a. x = 3

b. x = -3 and x = 3

c. x = 1 and x = 5

d. x = 1

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The graph of the function $f(x)$ is shown below. What is the value of the definite integral $\int_{-2}^{2} f(x)\,dx$?

a. 0

b. 2

c. 4

d. 8

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the function $f(x)$ whose derivative is given by $f'(x) = \frac{\sin(x)}{x^2+1}$. A related function $g(x)$ is defined by $g(x) = \int_0^x f(t)dt$. If $f(0) = k$, where k is a positive constant, find the value of $g'(0)$ and determine if $g(x)$ is increasing or decreasing at $x = 0$.

a. $g'(0) = k$, and $g(x)$ is increasing.

b. $g'(0) = 0$, and $g(x)$ is neither increasing nor decreasing.

c. $g'(0) = 1$, and $g(x)$ is increasing.

d. $g'(0) = k$, and $g(x)$ is decreasing.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Separable Equations

An AP Calculus BC FRQ models a population $P(t)$ satisfying the logistic differential equation $\frac{dP}{dt} = \frac{3}{600}P(600 - P)$. At what population size P is the population growing most rapidly?

a. $P = 300$

b. $P = 600$

c. $P = 150$

d. $P = 0$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

On an AP Calculus BC Free-Response Question, a particle moves along a curve with velocity vector $\vec{v}(t) = \langle 2t, 3t^2 \rangle$. What is the speed of the particle at time $t = 1$?

a. $\sqrt{13}$

b. 5

c. $\sqrt{5}$

d. 13

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

In a Taylor series BC FRQ, the alternating series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^3}$ is approximated by its first 2 terms S_2. According to the Alternating Series Error Bound, what is the maximum possible error $|S - S_2|$?

a. $\frac{1}{3^3} = \frac{1}{27}$

b. $\frac{1}{2^3} = \frac{1}{8}$

c. $\frac{1}{3^2} = \frac{1}{9}$

d. $\frac{1}{2 \cdot 3^3}$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Variation of Parameters

Consider the differential equation $y'' - xy = 0$. The point $x = 0$ is an ordinary point. If we assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^n$, find the recurrence relation for the coefficients c_n.

a. $c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$

b. $c_{n+2} = \frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

c. $c_{n+2} = -\frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$

d. $c_{n+2} = -\frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Separable Equations

Consider the system of linear differential equations: $\frac{dx}{dt}, = 3x + y; \frac{dy}{dt}, = x + 3y$ Find the eigenvalues of the coefficient matrix for this system.

a. $\lambda_1 = 2, \lambda_2 = 4$

b. $\lambda_1 = 1, \lambda_2 = 5$

c. $\lambda_1 = 3, \lambda_2 = 3$

d. $\lambda_1 = -2, \lambda_2 = 4$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Consider the differential equation $y''(t) + 4y(t) = u(t-2)$, where $u(t)$ is the Heaviside step function, with initial conditions $y(0) = 0$ and $y'(0) = 0$. Apply the Laplace Transform to both sides of the equation. If $Y(s) = \mathcal{L}\{y(t)\}$ and $\mathcal{L}\{u(t-a)\} = \frac{e^{-as}}{s}$, which of the following expressions represents $Y(s)$?

a. $\frac{e^{-2s}}{s(s^2+4)}$

b. $\frac{1}{(s^2+4)}$

c. $\frac{1}{s(s^2+4)}$

d. $\frac{e^{-2s}}{s^2+4}$

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. Use the Euler method with a step size $h = 0.1$ to approximate the value of $y(0.2)$.

a. 1.20

b. 1.21

c. 1.23

d. 1.24

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Variation of Parameters

Consider the ordinary differential equation $\frac{dy}{dx} = x + y$, with the initial condition $y(0) = 1$. We will use Euler's method to approximate the value of $y(0.2)$ using a step size of $h = 0.1$. Let $y_0 = y(x_0)$, $y_1 = y(x_1)$, and $y_2 = y(x_2)$ be the approximations at $x_0 = 0$, $x_1 = 0.1$, and $x_2 = 0.2$ respectively. Which of the following statements is TRUE regarding these approximations?

a. $y_1 = 1.1$ and the approximation for $y(0.2)$ is 1.21

b. $y_1 = 1.1$ and the approximation for $y(0.2)$ is 1.22

c. $y_1 = 1.2$ and the approximation for $y(0.2)$ is 1.32

d. $y_1 = 1.2$ and the approximation for $y(0.2)$ is 1.30

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